Feasibility Analysis of Airship-Deployed Hydrothermal Carbonization for Disaster Zone Waste Valorization

Technical Report

ME 401: Engineering Systems and Applications

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This report presents a computational feasibility analysis of a modular hydrothermal carbonization system conveyed by heavy-lift airship for on-site waste valorization in disaster-affected zones. Analysis is conducted as an exploration of a novel composite technology concept.

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EXECUTIVE SUMMARY

This report presents a computational feasibility analysis of a system concept combining heavy-lift airship logistics with modular hydrothermal carbonization (HTC) reactors for waste valorization in disaster-affected zones. The system is intended to operate without fixed infrastructure, converting locally generated organic waste into hydrochar, recoverable process heat, and treated process water while remaining energy self-sufficient.

The analysis is organized into three coupled modules. Module 3 establishes reactor vessel mass as a function of operating temperature and vessel geometry via hoop stress analysis. Module 1 translates vessel mass into a deployable module count and total system throughput, bounded by the Flying Whales LCA60T payload specification of 60,000 kg. Module 2 computes the gross thermal energy balance across a sweep of feedstock moisture content, mass flow rate, and reactor temperature. The primary deliverable is a throughput feasibility surface characterizing the design space available to the system under its combined structural, payload, and thermodynamic constraints.

1. INTRODUCTION

1.1 Motivation and Problem Context

Disaster events generate large, concentrated volumes of heterogeneous organic waste — municipal food waste, sewage sludge, agricultural residues, and contaminated soils — in zones where conventional waste management infrastructure has been damaged or destroyed. This waste poses immediate public health risks and impedes recovery operations, yet it also represents a latent energy and material resource.

Hydrothermal carbonization (HTC) is a thermochemical process that converts wet organic feedstock into hydrochar, process water, and recoverable gases at moderate temperatures (180–250°C) and autogenous pressures (10–40 bar). Unlike combustion-based approaches, HTC tolerates high-moisture feedstock without energy-intensive pre-drying, making it particularly well-suited to the wet, heterogeneous waste streams typical of disaster scenarios. The hydrochar product can be combusted as a fuel, applied as a soil amendment, or sequestered for carbon credit, depending on operational priorities.

The logistical challenge in disaster relief is reaching affected zones rapidly, before road and port infrastructure are restored. Heavy-lift airships offer a complementary deployment pathway: they require no prepared runway, can hover and lower cargo directly to ground-level sites, and are capable of carrying payloads on the order of tens of thousands of kilograms over ranges relevant to regional disaster response. The concept explored in this report combines these two technologies — airship conveyance of modular HTC reactor units — as a means of on-site waste valorization without reliance on fixed infrastructure.

This is a concept-stage feasibility study. The goal is not to produce a production-ready design, but to determine whether the composite technology concept is physically plausible within realistic payload, structural, and thermodynamic constraints, and to identify the key design tensions that would govern any further development.

1.2 Technical Objectives

- Determine reactor vessel mass as a function of operating temperature and vessel geometry using pressure vessel hoop stress analysis.
- Translate vessel mass and airship payload capacity into a feasible range of deployable module counts and total system throughput.
- Compute the gross thermal energy balance of the HTC process across the feasible throughput envelope and a range of feedstock moisture fractions.

- Identify the crossover moisture threshold above which preheating demand exceeds recoverable reaction energy, rendering a feedstock an energy liability.
- Assess how the coupled structural–payload–thermal constraints shape the viable operating envelope for this system concept.

2. SYSTEM DESCRIPTION

2.1 Concept Overview

The system consists of modular HTC reactor units sized for airship transport. A heavy-lift airship — the Flying Whales LCA60T, rated at 60,000 kg payload — carries multiple reactor modules to the deployment zone, where they are lowered to ground level for operation. Each module is a self-contained pressure vessel with associated preprocessing, separation, and energy recovery equipment. The number of modules per deployment is determined by the airship payload budget and individual module mass.

2.2 Process Chain

The HTC process is modeled as a five-stage chain:

1. **Intake and Preprocessing:** Waste reception, size reduction, and moisture conditioning. Energy cost of conditioning is bounded by feedstock energy value.
2. **HTC Reactor:** Thermochemical conversion at 180–250°C under autogenous pressure. Feedstock is converted to hydrochar, process water, and recoverable gas.
3. **Product Separation:** Phase separation of hydrochar, process water, and gas. Parasitic electrical load minimized.
4. **Energy Recovery:** Counterflow heat exchanger transfers heat from hot reactor outlet to incoming cold feed, reducing net preheat demand.
5. **Output Disposition:** Hydrochar is directed to fuel combustion or carbon sequestration depending on operational priority; process water is treated and recirculated; gas is handled or vented.

2.3 Energy Self-Sufficiency Condition

The central design requirement is energy self-sufficiency: the system must generate enough recoverable thermal energy from waste conversion to power its own operations. This is expressed as the governing inequality:

$$Q_{\text{net}}(\dot{m}, x_m, T_r, \varepsilon) \geq W_{\text{pump}} + W_{\text{aux}} \tag{1}$$

where Q_{net} is the net thermal power available after reaction, preheat demand, and heat exchanger recovery; W_{pump} is the pressurization work required to bring feed slurry to reactor pressure; and W_{aux} covers instrumentation, controls, and communications load. The present analysis evaluates Q_{net} on a gross basis (without heat exchanger recovery) as a conservative lower bound on system viability.

3. METHODOLOGY

3.1 Computational Architecture

The analysis is decomposed into three Python modules, implemented sequentially and linked through shared variables. This modular structure was chosen for two reasons: it makes each calculation independently verifiable, and it makes the coupling between structural, payload, and thermal constraints explicit rather than buried in a monolithic script.

Module 3 is implemented first because it is algebraically self-contained and produces the vessel mass that Module 1 requires. Module 1 is implemented second, consuming vessel mass and producing the throughput envelope consumed by Module 2. Module 2 is implemented last, taking throughput from Module 1 and evaluating the thermal balance across the parameter space.

The shared design parameter propagated across all modules is reactor temperature T_r, which simultaneously sets autogenous pressure (via the water saturation curve), vessel wall thickness (via hoop stress), module mass (via vessel geometry), deployable module count (via payload budget), and reaction energy yield (via feedstock thermochemistry). Sweeping T_r therefore propagates through the entire analysis chain, and this cross-module sensitivity is a primary output of the work.

3.2 Module 3: Pressure Vessel Mechanical Analysis

The HTC reactor operates under autogenous pressure — the vessel seals and pressurizes to the saturation pressure of water at the operating temperature. At 220°C, this corresponds to approximately 23 bar; at 250°C, approximately 40 bar. The required vessel wall thickness is governed by the thin-wall hoop stress relation:

$$t = \frac{P \cdot r}{\sigma_{\text{allow}} \cdot \eta_{\text{weld}}} \tag{2}$$

where P is autogenous pressure (Pa), r is vessel inner radius (m), $\sigma_{\text{allow}} = \sigma_y/\text{SF}$ is the allowable stress, and η_{weld} is weld efficiency. Material is 316 stainless steel ($\sigma_y = 205$ MPa), safety factor 3.5 gives $\sigma_{\text{allow}} \approx 58.6$ MPa, and $\eta_{\text{weld}} = 0.85$ for radiographed welds.

Total vessel mass is computed as:

$$M_{\text{vessel}} = \rho_{\text{steel}} \cdot 2\pi r L t \cdot (1 + f_{\text{cap}}) \tag{3}$$

where vessel length $L = 3.0$ m and end-cap fraction $f_{\text{cap}} = 0.175$ approximates ellipsoidal head geometry. Autogenous pressure is extracted from the water saturation curve via CoolProp. The thin-wall validity condition $t/r < 0.1$ is checked at every operating point.

Temperature is swept over $T_r \in \{180, 200, 220, 250\}$°C and inner radius over $r \in [0.5, 1.0]$ m. The output is $M_{\text{vessel}}(T_r, r)$, saved to `module3_output.json`.

3.3 Module 1: Payload and Buoyancy Feasibility

Module 1 translates vessel mass into a deployable module count and total system throughput. Module mass is estimated by applying a scale factor of 1.5 to vessel mass, accounting for plumbing, internals, and structural framing:

$$M_{\text{module}} = 1.5 \cdot M_{\text{vessel}} \tag{4}$$

Module count is then:

$$N = \left\lfloor \frac{M_{\text{payload}}}{M_{\text{module}}} \right\rfloor \tag{5}$$

where $M_{\text{payload}} = 60{,}000$ kg (Flying Whales LCA60T rated capacity). Total system mass flow rate is computed from reactor geometry and residence time:

$$\dot{m}_{\text{total}} = N \cdot \frac{\rho_{\text{slurry}} \cdot V_{\text{reactor}}}{\tau} \tag{6}$$

where slurry density $\rho_{\text{slurry}} = 1000$ kg/m^3, reactor volume $V = \pi r^2 L$, and residence time τ is swept over $\{1, 2, 4\}$ hours, covering the HTC literature range for municipal feedstocks. Fill fraction is assumed 1.0 as an upper bound. Design points where $N = 0$ (vessel mass exceeds payload) are flagged and excluded.

A key finding from Module 1 is that total fleet throughput does not increase monotonically with vessel radius. The $\lfloor \cdot \rfloor$ discretization of module count produces a sawtooth pattern as radius increases: volume grows as r^2 while module count drops in integer steps. The optimal deployment radius is the local maximum of $N \cdot \dot{m}$ for each (T_r, τ) combination.

3.4 Module 2: Reactor Thermal Energy Balance

Module 2 computes the gross thermal energy balance of the HTC process. The governing equations are:

$$Q_{\text{rxn}} = \dot{m} \cdot \text{HHV}_{\text{dry}} \cdot \eta_{\text{HTC}} \cdot (1 - x_m) \tag{7}$$

$$Q_{\text{pre}} = \dot{m} \cdot x_m \cdot c_{p,w} \cdot (T_r - T_{\text{feed}}) \tag{8}$$

$$Q_{\text{net}} = Q_{\text{rxn}} - Q_{\text{pre}} \tag{9}$$

where x_m is moisture fraction (swept 0–1), HHV_{dry} is the higher heating value of the dry feedstock fraction, η_{HTC} is the HTC energy conversion efficiency, $c_{p,w} = 4180$ J/kg·K, and $T_{\text{feed}} = 20°$C.

Two feedstocks are modeled as representative of disaster municipal waste: municipal food waste (HHV_{dry} = 16 MJ/kg, η_{HTC} = 0.60) and sewage sludge (HHV_{dry} = 12 MJ/kg, η_{HTC} = 0.55). These values are placeholders pending replacement with peer-reviewed literature values (Kambo & Dutta, 2015; Funke & Ziegler, 2010).

$Q_{\text{net}} > 0$ indicates a thermally self-sustaining operating point on a gross basis. The crossover moisture threshold x_m^* where $Q_{\text{net}} = 0$ is the primary scalar output per feedstock and operating condition. The full output is a surface $Q_{\text{net}}(\dot{m}, x_m, T_r)$ saved to `module2_output.json`.

4. RESULTS AND DISCUSSION

4.1 Pressure Vessel Mass and Structural Constraint

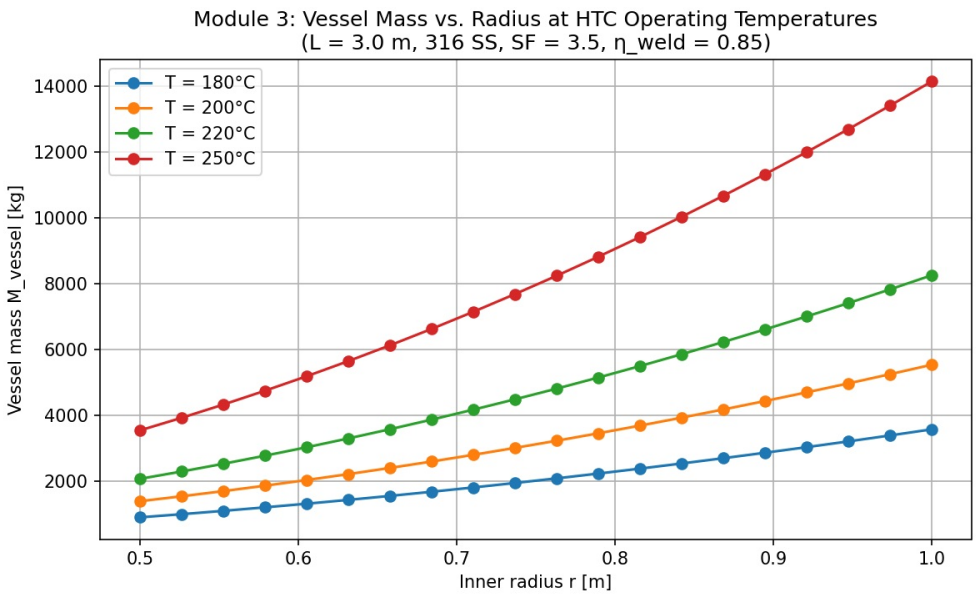

Figure 1: Reactor pressure vessel mass as a function of inner radius at four operating temperatures. Higher temperatures produce greater autogenous pressure, requiring thicker walls and heavier vessels. Vessel length fixed at $L = 3.0$ m; material 316SS with safety factor 3.5.

Reactor vessel mass increases with both operating temperature and vessel radius, as expected from the hoop stress relation (Eq. 2). The dominant driver of mass growth with temperature is the non-linear pressure–temperature relationship along the water saturation curve: pressure approximately doubles between 200°C and 250°C, producing a corresponding increase in required wall thickness and vessel mass. This has direct downstream consequences for deployable module count.

4.2 Throughput Feasibility Surface

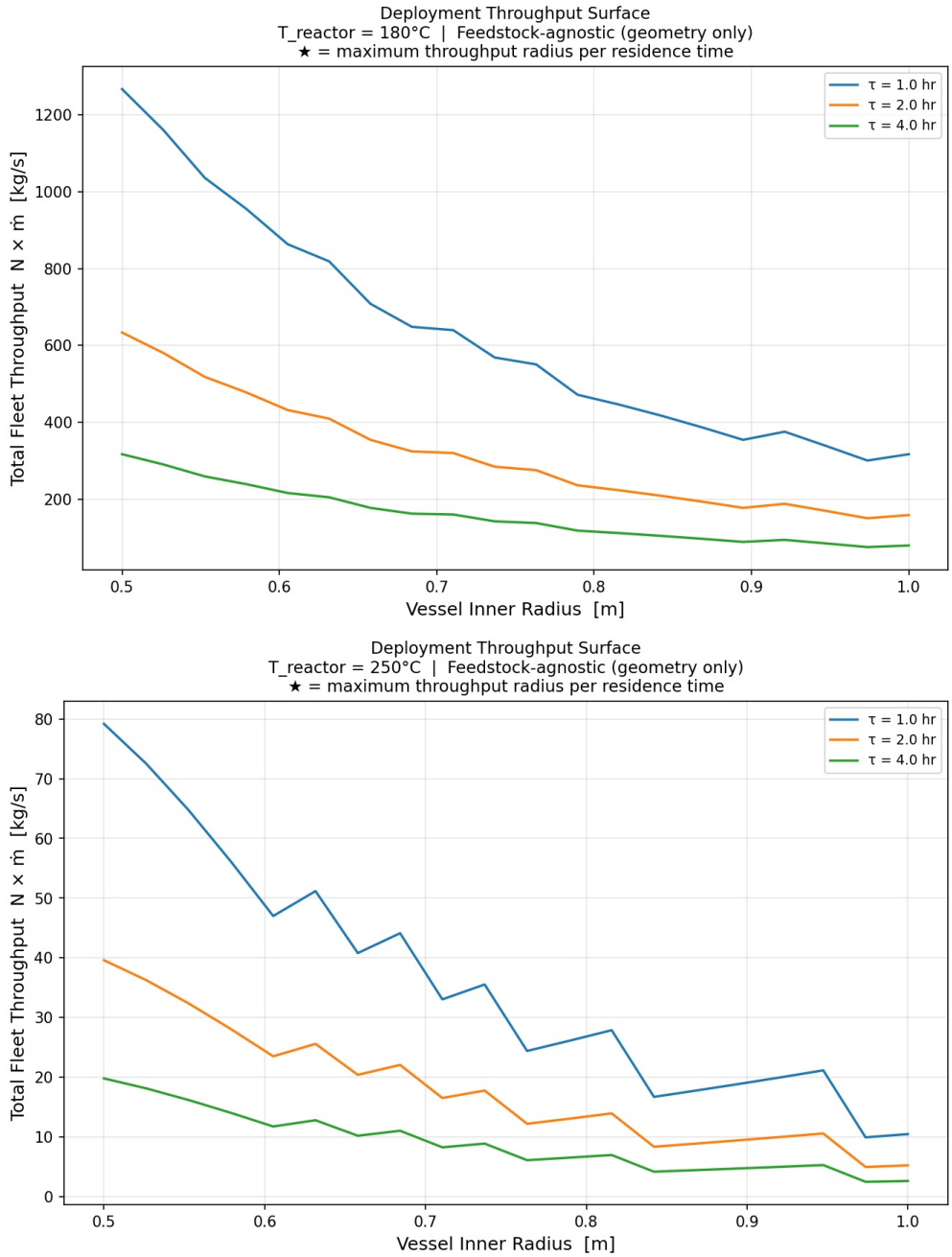

Figure 2: Total fleet throughput ($N \cdot \dot{m}_{\text{total}}$) as a function of vessel inner radius, parameterized by reactor temperature and residence time. Demonstrating upper and lower boundaries of throughput. The sawtooth profile reflects integer discretization of module count. Stars not in figure title not included (removed from code but not heading).

Total fleet throughput exhibits a non-monotonic dependence on vessel radius due to the integer floor constraint on module count (Eq. 5). As radius increases, reactor volume grows as r^2, in-

creasing per-module throughput capacity. However, module mass grows faster, and at each radius where module count drops by one integer, throughput falls discontinuously. The envelope of local maxima defines the Pareto-optimal set of vessel radii for a given temperature and residence time.

The optimal radius appears dependent of residence time, suggesting the geometric and payload constraints dominate over process kinetics in determining the best module size at this stage of analysis. The stronger sensitivity is to T_r: higher temperatures reduce module count and compress the feasible throughput range.

4.3 Gross Thermal Energy Balance

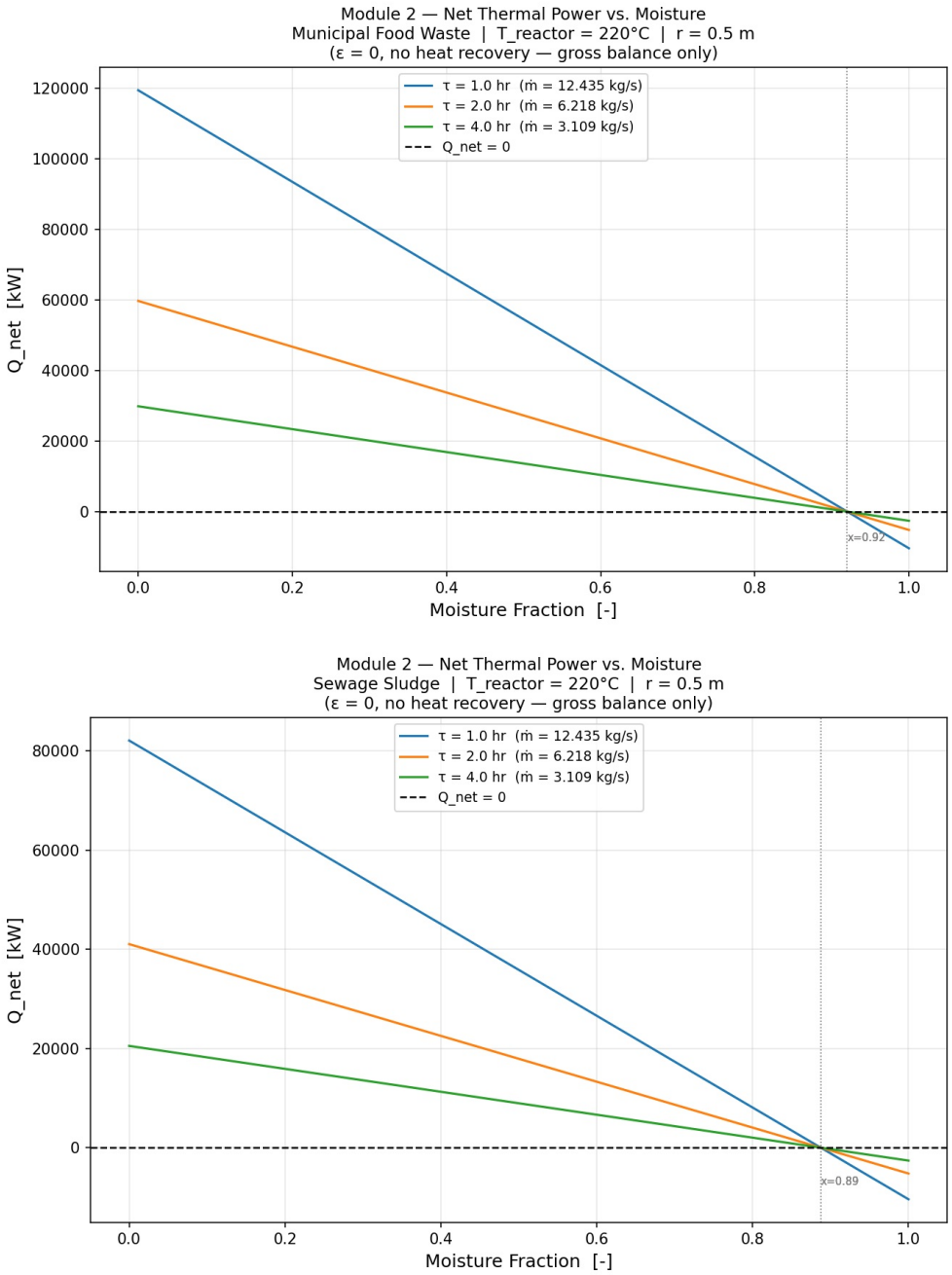

Figure 3: Gross net thermal power Q_{net} as a function of feedstock moisture fraction for municipal food waste and sewage sludge, at representative throughput values from Module 1. The crossover moisture threshold x_m^* (where $Q_{\text{net}} = 0$) marks the upper bound of thermally viable operation on a gross basis.

Q_{net} decreases linearly with moisture fraction for both feedstocks, reflecting the competition between reaction energy release (proportional to dry mass fraction) and preheat demand (proportional to water mass fraction). The crossover threshold x_m^* represents the moisture fraction above which the system cannot sustain itself on gross thermal output alone. Municipal food waste, with a higher HHV and conversion efficiency, tolerates higher moisture fractions than sewage sludge before crossing into energy deficit.

It is important to note that this is a gross balance: heat exchanger recovery (the ε term in the full self-sufficiency inequality, Eq. 1) will shift x_m^* upward by reducing net preheat demand. The gross crossover threshold is therefore a conservative lower bound on operable moisture range, not the operational limit of the fully integrated system.

4.4 Coupling Between Modules: The Temperature Tension

The most important result of this analysis is not contained in any single module but in the coupling between them. Reactor operating temperature T_r enters all three modules simultaneously and creates competing effects:

- **Module 3:** Higher $T_r \rightarrow$ higher autogenous pressure $\rightarrow$ thicker walls $\rightarrow$ heavier vessels.
- **Module 1:** Heavier vessels $\rightarrow$ fewer modules per deployment $\rightarrow$ lower total throughput.
- **Module 2:** Higher $T_r \rightarrow$ greater preheat demand $\rightarrow$ lower Q_{net} at fixed moisture; but also faster reaction kinetics (not yet modeled explicitly) suggesting shorter residence times may be viable, recovering some throughput.

There is therefore a non-trivial optimal temperature that balances structural mass penalty against thermochemical performance. Identifying this optimum is the natural next step for this analysis, requiring explicit integration of kinetic rate data with the existing thermal and mechanical models.

5. LIMITATIONS AND DEFERRED WORK

Several modeling assumptions warrant explicit acknowledgment:

Placeholder thermochemical values. HHV_{dry} and η_{HTC} for both feedstocks are first-pass estimates. These values drive Q_{net} directly and must be replaced with peer-reviewed literature values (Kambo & Dutta, 2015; Funke & Ziegler, 2010) before conclusions about self-sufficiency margins are quantitatively reliable.

Scale factor on module mass. The $1.5\times$ multiplier on vessel mass to estimate total module mass is engineering judgment. A bill of materials accounting for pumps, heat exchangers, instrumentation, and structural framing is needed to refine this estimate.

Heat exchanger recovery excluded. Module 2 computes a gross balance without heat exchanger effectiveness ε. The full self-sufficiency inequality (Module 5 in the original project scope) would apply ε to reduce net preheat demand, shifting crossover thresholds upward. This is the most significant near-term extension.

Pump and auxiliary work excluded. W_{pump} and W_{aux} are not subtracted in the current balance. These are parasitic loads that work against self-sufficiency and should be included in any refined analysis.

Volumetric payload check not performed. Module 1 applies only a mass constraint (60,000 kg). Whether N modules physically fit within the cargo bay volume of the LCA60T has not been verified.

Vessel length fixed. $L = 3.0$ m is a placeholder. Vessel length should be derived from the required reactor volume at each (T_r, r, τ) combination once residence time is explicitly coupled to reaction kinetics.

6. FURTHER IDEAS AND FUTURE DIRECTIONS

This analysis establishes a structural and thermodynamic envelope for the system concept. Several extensions are of natural engineering interest:

Lifting gas as supplemental fuel. The LCA60T uses hydrogen as its lifting gas. As reactor modules are lowered and airship mass decreases, the vessel becomes more buoyant. Selectively venting hydrogen at this point and combusting it as supplemental reactor fuel represents an operationally convenient energy source (H_2 LHV ≈ 120 MJ/kg). The tradeoff — energy gained vs. lift capacity sacrificed for subsequent operations — is a coupled buoyancy–energy calculation that could be appended to Module 1.

Methane recovery from HTC process gas. HTC produces a gas phase containing CO_2, CO, and methane. Recovering and combusting the methane fraction would supplement reactor energy input and improve the self-sufficiency margin. Quantifying this contribution requires gas yield data from HTC literature as a function of feedstock and temperature.

Hydrochar as carbon sequestration product. If energy self-sufficiency can be achieved without combusting the hydrochar, the char can be applied as a soil amendment or formally sequestered for carbon credit. This changes the system's optimization objective from energy output to carbon removal, which may be more valuable in certain disaster recovery contexts (e.g., agricultural zone restoration). The strategic choice between fuel and sequestration is a design decision that the energy balance analysis directly informs.

Adaptive temperature control for mixed feedstocks. Disaster waste is heterogeneous. A single fixed T_r may not be optimal across the range of feedstocks encountered in the field. An adaptive operating logic that adjusts temperature in response to estimated feedstock moisture and heating value would require a feedback control layer on top of the existing steady-state modules.

Full self-sufficiency integration (Module 5). The original project scope defined a Module 5 that assembles Q_{net}, W_{pump}, and W_{aux} into a two-dimensional feasibility surface in $\{\text{moisture} \times \text{throughput}\}$ space. Implementing this integration would produce the terminal deliverable of the full analysis: a quantitative map of the self-sufficient operating region.

7. AI-ASSISTED ENGINEERING: PROCESS AND REFLECTION

7.1 Computational Development Strategy

Claude (Anthropic) was used as a collaborative brainstorming partner throughout the scoping and design phases of this project. The primary workflow was concept-first: before any code was written, effort was invested in building clear, structured documentation of the system concept, process chain, and analytical scope. This documentation was developed iteratively through conversation with the model and maintained as a version-controlled context file passed into each session.

7.2 Lessons Learned in AI-Augmented Engineering

A key observation from this process was that Claude's misinterpretation of early prompts served as a diagnostic tool. Where the model produced unexpected outputs or asked clarifying questions, this reliably identified gaps in the problem framing rather than gaps in the model's capability. Each such instance prompted a refinement of the context document, progressively sharpening the problem definition.

The structured scoping work culminated in a literal pseudocode artifact detailing the module sequence, governing equations, shared variables, and coupling logic. With this artifact in place, the transition to working Python code was rapid — the full three-module implementation was

completed in under one hour with no significant troubleshooting. The investment in upfront clarity directly compressed the implementation phase.

8. CONCLUSIONS

8.1 Key Findings

- Reactor vessel mass is strongly sensitive to operating temperature, driven by the nonlinear pressure increase along the water saturation curve. This creates a direct structural penalty for higher-temperature operation that propagates into reduced deployable module count.
- Total fleet throughput is non-monotonic in vessel radius due to integer discretization of module count. An optimal deployment radius exists for each operating condition and is largely insensitive to residence time, indicating geometric and payload constraints dominate at this analysis stage.
- Gross thermal self-sufficiency is feasible for both municipal food waste and sewage sludge feedstocks below characteristic crossover moisture thresholds. Food waste tolerates higher moisture fractions due to its greater HHV and conversion efficiency.
- The coupled analysis reveals a temperature tradeoff that cannot be resolved within any single module: higher T_r improves reaction kinetics but increases structural mass and reduces deployable capacity. Resolving this tradeoff is the central open question for future work.

8.2 Recommendations for Further Development

The analysis supports continued development of the concept with the following near-term priorities: (1) replace placeholder thermochemical values with peer-reviewed HTC literature data to establish quantitatively reliable self-sufficiency margins; (2) implement heat exchanger recovery (Module 4/5) to determine the practical moisture operating range; (3) conduct a volumetric payload check against the LCA60T cargo bay geometry; and (4) evaluate the hydrogen offtake energy contribution as a potential means of extending the operable moisture range or reducing required throughput for self-sufficiency.

The system concept is physically plausible within the constraints examined. Whether it is operationally viable depends on factors beyond the current scope: deployment logistics, safety and regulatory frameworks for airborne chemical processing, and the economic comparison against alternative disaster waste management approaches. These represent the appropriate scope of a follow-on study.